My agent workbench

Making things with AI.
Keeping enough evidence to trust them.



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AI & Ales · Jamestown Frontier Society
October 15, 2026 · Billsburg Brewery

The tools around my agent

Codex	A working partner in my projects, files and terminal
Skills + Markdown	Reusable instructions and decisions that survive a chat
Git + worktrees	Separate experiments, inspect changes, recover a state
Pixi + DataLad	Recover environments and connect data to the work

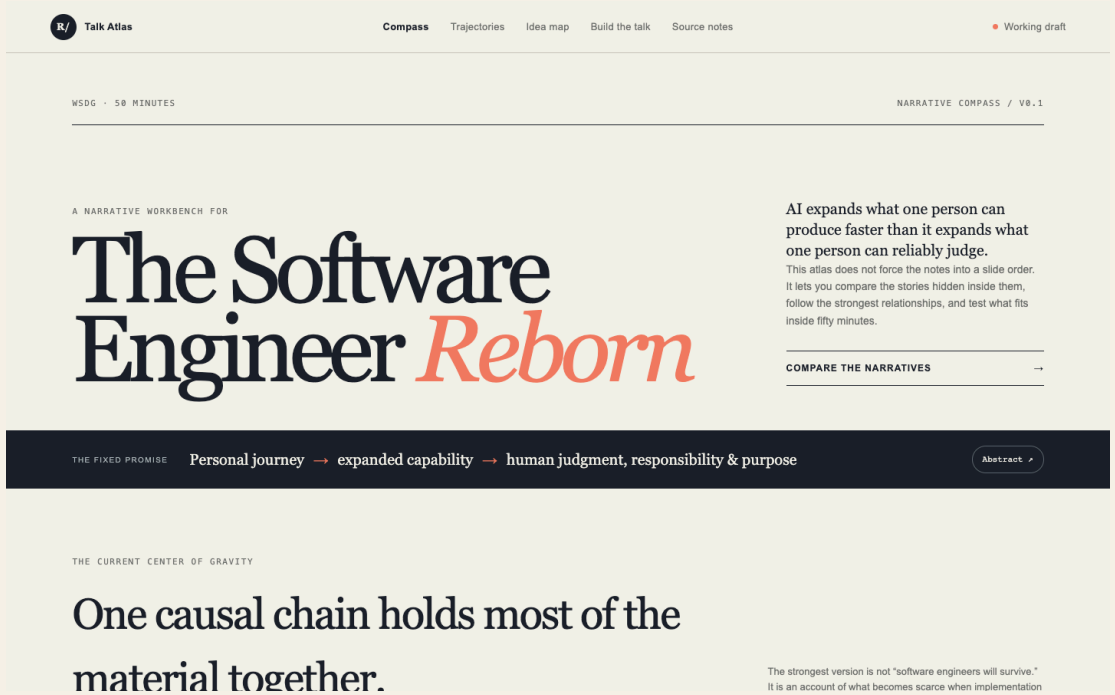
My skill workshop

Worktree tools

Execution records: duct

The Software Engineer Reborn

I used an interactive preparation space to work through the ideas.



Explore the preparation



Open the presentation

What did we just make?



A research object includes the result and what it takes to inspect or rerun it.

STAMPED asks about the package: its history, its parts, its environment, and how someone else can use it.

Research objects with AI: the talk

STAMPED principles

Orinoco Lite

Connected research information

People, projects and outputs share explicit relationships.

An editor proposes changes.
GitHub review preserves the history.
The records feed a website and graph.



Project + source

Explore the graph

Poster

STAMPED in Practice

Skills that improve through scientific work

Study record on Zenodo

Next: run, assess, improve, repeat

Compare four assistance conditions.
Test quality, effort and consistency.



STAMPED · US-RSE 2026

STAMPED in Practice

Checking AI-assisted analysis and its research record

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Poster + source

An AI assistant can write analysis code. **Can another person check its result, rerun the work, and reuse what it produced?** STAMPED gives us seven practical questions to ask about that research.

We compare the same tasks with and without five reusable skills.

A skill is a short set of instructions and checks an assistant can follow. We check scientific results and the records needed to reproduce them.

1 · WHAT WOULD MAKE THIS WORK REUSABLE?

Seven properties, with evidence for each

Three paired tasks show useful evidence and limits. A recorded command alone does not show that an analysis is correct.

Property	Practical question	Original abstract: tool plan	Three pairs: evidence and gaps
Self-contained	Is everything needed identified?	DataLad + git-annex	Partial · both arms Pinned inputs and locked software; offline setup untested.
Trackable	Can we follow how it changed?	Git · DataLad run · con-duct · NIDM · PROV	Partial · both arms Git and row audits retained; native DataLad rerun incomplete.
Actionable	Can someone run or use it?	Tested BIDS Apps + Pixi tasks	Demonstrated · both arms All six science commands replayed; full reports untested.
Modular	Can a part be reused on its own?	DataLad composition	Demonstrated · both arms Three supplied datasets; workers separated code and outputs.
Portable	Can it run in another environment?	Apptainer	Partial · both arms New software environments, one Mac; other systems untested.
Ephemeral	Can a fresh workspace reproduce it?	Apptainer · BABS · Slurm	Partial · both arms Outputs rebuilt in new checkouts; shared caches retained.
Distributable	Can others obtain the right parts?	Persistent git-annex remotes	Partial · both arms Input license known; new code licensing and external retrieval unverified.

Tools preserve the original proposal; many were not used. Judgments cover these three pairs. The coordinator supplied datasets and ran independent replay; later log packaging is curation. Containers and clusters are untested.

2 · A SMALL, INCREASINGLY DEMANDING CASE STUDY

How do response times change across runs?

Study + evidence

We use public mixed-gambles task records from OpenNeuro ds000005: 16 participants and three runs each. We summarize recorded decision times; we do not infer why they changed.

01 One participant
Compare the median response time in the first and third runs.

02 The group
Estimate the average within-person change and its uncertainty.

03 Sensitivity
Vary the summary and response-time rule; add the middle run.

Each stage uses two fresh assistant contexts. Progress depends on a correct result and an independent successful replay at the preceding stage.

Checked scientific results

One participant: median response time 1272 → 1261 ms; change -11 ms. Both arms agree.

16 participants: average within-person change +31.3 ms; 95% bootstrap interval [-70.3, 132.7] ms. Both arms agree.

Change is third minus first run; the interval includes zero. Missing responses limit interpretation. This is not a causal effect.

Sensitivity: median/mean summaries, with or without a 0.2–3.0 s response-time band, gave +11.2 to +58.7 ms; all four 95% intervals include zero.

Scope change: public behavioral data replace the proposed ABCD neuroimaging case. **The ABCD result has not been reproduced.**

3 · WHAT DO THE SKILLS ADD?

A matched comparison

Both arms receive the same data, task, software, STAMPED references, and shared account controls. Only the treatment arm is instructed to read the five new skills.

Baseline
Common task and controls

With skills
Common task + five skills

Each attempt: 8 minutes · gpt-5-astra · ultra-reasoning · fresh local workspace.

Reusable research guidance

The skills turn the seven properties into steps: define the research object, preserve evidence, make work runnable, check it, and prepare it for reuse.

Read the skills

REPORT BENEFITS AND COSTS TOGETHER

What do we check?

- **Data checks:** separate numerical and row-level checks.
- **Replay:** remove outputs, then rebuild in a fresh checkout.
- **Costs and gaps:** time, tokens, setup, interventions, and failures.

ONE ATTEMPT PER ARM AND TASK

Checked outcomes

Task	Baseline	With skills
One person	270 s Data checks: Passed Replay: Passed	358 s Data checks: Passed Replay: Passed
Cohort	387 s Data checks: Passed Replay: Passed	413 s Data checks: Passed Replay: Passed
Sensitivity	383 s Data checks: Passed Replay: Passed Screening: Passed	474 s Data checks: Passed Replay: Passed Screening: Failed

Worker time only; setup and evaluator replay are separate. Replay used a fresh checkout on the same host.

Screening and integration failures

The skills arm in task 03 failed the required case-insensitive "N/A" check. Those tokens were absent from the real data. Missing validation software and native DataLad rerun also caused failures.

Limits of this study

All trials share an account, host, and software caches. Fresh contexts do not isolate account settings.

An unblinded study-design agent checked the outputs; the authors develop STAMPED and these skills. This small case series cannot establish general effectiveness.

Three local pairs, October 2026. Source: Tom et al. (2007), mixed-gambles task, OpenNeuro ds000005 (PDDL). Code, input versions, protocol, and review notes are linked from the study.

STAMPED means Self-contained, Trackable, Actionable, Modular, Portable, Ephemeral, Distributable. Poster layout adapted from CON's US-RSE 2026 ORINOCO Lite poster.

OpenAI Codex assisted the analyses and poster draft.

Explore CON

Dartmouth College Psychological and Brain Sciences

Supported by ReproNim

Research workshop in development.
Transfer remains to be tested.

Poster PR #35

Study on Zenodo

Next repository

What would convince you?

What evidence do you need before you trust something an agent made?

I’m exploring how to make that evidence easier to keep and inspect.

leej3.github.io/ai-and-ales-2026

All links, downloads and editable source